NTWR Prime - redundant security based on NTRU Prime and LWR problems

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Abstract In this article, we introduce a new post-quantum cryptosystem, *NTWR Prime*, which is based on the NTRU Prime and Learning With Rounding (LWR) problems. This scheme is inspired by the NTWE construction proposed by Joel Gärtner in 2023 [1]. Unlike NTWE, our algorithm employs an irreducible, non-cyclotomic polynomial whose Galois group is isomorphic to the symmetric group. Additionally, the LWR problem is used in place of the LWE problem, offering potential advantages for structural security due to its deterministic nature.

We conduct a security analysis demonstrating that solving the NTWR Prime problem requires solving both the underlying NTRU Prime and LWR problems. Consequently, given the absence of definitive post-quantum security proofs for these problems, our construction offers redundancy, which may fulfill the requirements of applications with exceptionally high security standards. Importantly, we show that there exists a set of parameters satisfying the hardness assumptions for both contributing problems.

Keywords: PQC · NTRU Prime · LWR.

1 Introduction

Algorithms based on lattices currently represent one of the main approaches to post-quantum cryptography (PQC) [2]. This is due to the fact that lattices allow for the definition of problems for which NP-hardness proofs are available [3]. This applies, in particular, to the Shortest Vector Problem (SVP) on lattices and the Closest Vector Problem (CVP). The NP-hardness of these problems makes them infeasible to solve efficiently in polynomial time. Importantly, this holds true for both classical and quantum computation. The class of NP-hard problems is believed to be not contained within the BQP (Bounded Quantum Polynomial) class, which includes problems solvable in polynomial time by quantum computers (e.g., the factoring problem). Therefore, NP-hard problems are a promising direction for the development of post-quantum algorithms.

The prototype of lattice-based algorithms can be considered to be the NTRU algorithm, introduced in 1996 [4]. This algorithm is based on the quotient ring of polynomials $R = \mathbb{Z}[x]/(x^N - 1)$, where N is chosen to be a sufficiently large prime number (e.g., 401, 509, 677, 821). Additionally, the NTRU cryptosystem uses two relatively prime integers q and p, such that q is much larger than p (neither p nor q need to be prime), and p should be odd. An example of such a pair is $q = 2048$ and $p = 3$.

The private and public key pair is given by:

$$(f, g) \in R \times R, \quad h = f^{-1} \star g \mod q \in R, \tag{1}$$

where f and g are secret polynomials with appropriately small coefficients, generated from a specified distribution.

The idea of the connection between NTRU and the lattice problem is based on the use of lattices to perform an attack on the cryptosystem. Specifically, we can define a lattice as follows:

$$\mathcal{L}_h = \{(a, b) \in \mathbb{Z}^{2N} : a \star h = b \mod q\},$$

where h is the public key. By the definition of h, it holds that $f \star h = g \mod q$. Therefore, the private key (f, g) is an element (a vector) in the lattice $\mathcal{L}_h$.

Since $f \star h - g = qt$ for some integer vector t, we can express the vector (f, g) as follows: $(f, g) = (f, -t)M$, where the matrix M is given by:

$$M = \begin{pmatrix} I & H \\ 0 & qI \end{pmatrix}. \tag{2}$$

The submatrix H is formed from the cyclic permutations of the vector h. The rows of the matrix M form the basis vectors of the lattice $\mathcal{L}_h$.

Since the coefficients of the vector (f, g) are small, this vector is, with high probability, the shortest vector in $\mathcal{L}_h$. In this way, the security of the NTRU algorithm is related to the SVP problem, highlighting the NP-hardness of cryptanalyzing this algorithm.

In 2016, the article [5] proposed improvements to the NTRU algorithm in the form of the NTRU Prime algorithm. The main idea behind this modification concerned the polynomial $P(x) = x^N - 1$, which defines the quotient ring. The authors of the NTRU Prime cryptosystem pointed out potential vulnerabilities of the NTRU algorithm related to the cyclotomic nature and the symmetry of the roots o f this polynomial. As a remedy, they proposed the use of a quotient ring of polynomials lacking internal structure. This is achieved through the use of *non-cyclotomic polynomials* whose Galois group is isomorphic to a permutation group. Based on this, the NTRU Prime algorithm employs the ring $R = \mathbb{Z}[x]/(x^p - x - 1)$, where $p \in \mathbb{P}$, satisfying these requirements.

In 2005, Oded Regev introduced a new class of computationally hard problems, known as the *Learning With Errors* (LWE) problem [6]. Importantly, the LWE problem can be reduced to the approximate Shortest Vector Problem (SVP). However, there is no proof that it is NP-hard.

The idea behind LWE is based on the following observation. For a certain matrix A_{ij} and a vector b_i such that $b_i = A_{ij}s^j$, finding the vector s_j is easy (e.g., using Gaussian elimination). However, this problem becomes difficult after adding an error vector e_i: $b_i = A_{ij}s^j + e_i$.

The use of the above observation in constructing an asymmetric cryptosystem relies on the following construction of the private and public keys:

$$\bar{s} \in \mathbb{Z}_q^k, \quad (\bar{a}_i, b_j) = (\bar{a}_i, \langle \bar{a}_j, \bar{s} \rangle + e_j) \in \mathbb{Z}_q^k \times \mathbb{Z}_q,$$

where $i, j = 1, 2, \ldots, h$, and the scalar product $\langle \bar{a}_j, \bar{s} \rangle = (a_j)^i s_i = A_j{}^i s_i$. Alternatively, these keys can be written as:

$$(A, \bar{b} = A\bar{s} + \bar{e}) \in \mathbb{Z}_q^{h \times k} \times \mathbb{Z}_q^h. \tag{3}$$

We consider two main types of attacks on the LWE problem, namely the *Primal Lattice Attack* and the *Dual Lattice Attack* [7, 8, 9]. In the Primal Lattice Attack, we use the following lattice:

$$\mathcal{L} = \{v \in \mathbb{Z}^{h+k+1} : (A \mid I_h \mid -\bar{b})v = 0 \mod q\}. \tag{4}$$

As can be easily observed, the shortest vector in this lattice is $v = (\bar{s}, \bar{e}, 1)$, and thus, finding it allows one to recover the secret key $\bar{s}$.

In the case of the *Dual Lattice Attack*, we use the lattice:

$$\mathcal{L}' = \{(x, y) \in \mathbb{Z}^h \times \mathbb{Z}^k : A^T x = y \mod q\}.$$

A short vector (v, w) in this lattice allows the construction of a "distinguishing" value:

$$z = v^T \bar{b} = v^T (A\bar{s} + \bar{e}) = w^T \bar{s} + v^T \bar{e},$$

which is small and approximately follows a Gaussian distribution. This enables a distinguishing attack that differentiates between a Gaussian distribution and a uniform distribution for uniformly sampled $\bar{b}$.

In 2023, J. Gärtner proposed the NTWE cryptosystem [1], which synthesizes the NTRU cryptosystem with the LWE-based approach. It is worth noting that in Gärtner's proposed scheme, the polynomial ring $R = \mathbb{Z}[x]/(x^n + 1)$ is used, which does not possess a maximal symmetry group of roots. Therefore, it does not exhibit the characteristics typical of NTRU Prime.

Moreover, NTWE departs from the assumption that the functions f and g in NTRU come from the same distribution, as well as from the assumption that the coefficients of g are sufficiently small.

The private and public key pair in the NTWE algorithm is defined as:

$$\begin{aligned} &(f, \bar{s}) \in R \times R_q^k, \\ &(\bar{a}_i, b_j) = \left(\bar{a}_i, f^{-1} \star \underbrace{(\langle \bar{a}_j, \bar{s} \rangle + e_j)}_{=g} \right) \in R^k \times R, \end{aligned}$$

where $i, j = 1, 2, \ldots, h$.

The analysis of NTWE leads, among other things, to the conclusion that it may offer increased resistance to Dual Lattice Attacks and Primal Lattice Attacks compared to LWE. Based on current analysis, it can be conjectured that the complexity of the NTWE problem is no less than that of the NTRU and LWE problems considered separately. However, this issue requires further investigation.

It should be noted that the Learning With Errors (LWE) problem is a non-deterministic process, due to the use of a random error generator. This fact is significant from both a security and performance perspective. Specifically, in terms of security, the use of pseudorandom generators introduces an additional potential attack vector against the cryptosystem. It also increases the complexity of the algorithm, thereby raising the likelihood of implementation errors. Furthermore, repeated generation of pseudorandom numbers during the execution of the algorithm instance negatively impacts performance. These considerations also apply to the NTWE algorithm, which incorporates the LWE mechanism.

An alternative to LWE, which is deterministic in nature, is the *Learning With Rounding* (LWR) problem, proposed in 2011 [10]. In this problem, instead of adding Gaussian noise (as in LWE), a rounding function is applied, which makes the recovery of the secret message more difficult.

In the LWR-based approach, the private and public key pair is defined as:

$$\bar{s} \in \mathbb{Z}_q^k, \quad (\bar{a}_i, b_j) = (\bar{a}_i, \lfloor \langle \bar{a}_j, \bar{s} \rangle \rfloor_p) \in \mathbb{Z}_q^k \times \mathbb{Z}_p,$$

where for $x \in \mathbb{Z}_q$ and $q > p$, the rounding function is defined as $\lfloor x \rfloor_p := \left\lfloor \frac{xp}{q} \right\rfloor$.

This article presents a proposal for the NTWR Prime cryptosystem, which modifies the NTWE algorithm into a form based on the LWR problem. Furthermore, the cryptosystem proposed employs a non-cyclotomic polynomial whose Galois group is isomorphic to a permutation group. On this basis, NTWR Prime is built upon the ring $R = \mathbb{Z}[x]/(x^p - x - 1)$, as used in the NTRU Prime algorithm.

We therefore expect a reduction in the space of possible attacks on NTWR Prime compared to NTWE, both in terms of vulnerabilities related to the symmetries of the polynomial $P(x)$ (e.g., hindering meet-in-the-middle attacks), as well as in terms of the nondeterministic nature of NTWE. In the latter case, this especially concerns the simpler construction of the cryptosystem, which introduces higher "constructional security" due to a more straightforward implementation and, in particular, a reduced number of potential implementation errors. NTWR Prime, apart from the key generation phase, also does not require the use of randomness, the generation of which is both time-consuming and potentially introduces an additional attack vector.

It is worth emphasizing that NTWR Prime, like NTWE, embodies the concept of redundancy–a well-known approach to enhancing security by duplicating critical components of the system. In this case, security relies on the hardness of two problems: NTRU Prime and LWR. Since the security of both problems is not yet fully understood, a natural question arises: what happens if one of them is compromised? In our construction, security is still upheld by the remaining problem.

Naturally, the price paid for this elevated level of security is increased key size and reduced performance. Therefore, we stress that our proposal may not satisfy the efficiency requirements of standard, widely used applications. Nonetheless, the construction presented here may be a suitable candidate in scenarios where a high security margin takes precedence over efficiency.

The structure of the article is as follows. Section 2 introduces the NTWR Prome problem, which forms the basis of our contradiction. Both the decision and search variants of the problem are defined. Section 3 describes the generation of private and public keys. Building on this, Section 4 presents the encryption and decryption procedures, along with a proof of correctness for the decryption. A detailed security analysis of the proposed cryptosystem is provided in Section 5. Section 6 introduces exemplary parameter sets that ensure the hardness of the two underlying problems, along with an analysis of the decryption error. Finally, Section 7 summarises our findings and outlines directions for future research. The article concludes with an Appendix that presents a step-by-step, pedagogical toy implementation of the cryptosystem.

2 NTWR Prime Problem

We consider the rings $R_p = \mathbb{Z}_p[x]/P(x)$, $R_q = \mathbb{Z}_q[x]/P(x)$, and $R = \mathbb{Z}[x]/P(x)$, such that $p, q \in \mathbb{P}$ and $p < q$, with $R_p \subset R_q \subset R$. The polynomial

$$P(x) = x^n - x - 1, \tag{5}$$

for some values of n, p and q is irreducible and non-cyclotomic, whose Galois group is isomorphic to the symmetric group [11].

The irreducibility of the polynomial $P(x)$ guarantees the existence of an inverse for any non-zero polynomial f in $R_q = \mathbb{Z}_q[x]/P(x)$, which simplifies the key generation procedure. Specifically, the irreducibility of the polynomial (5) in the ring R_q implies that this ring becomes a finite field, which in turn ensures the existence of an inverse for every $f \in R_q \setminus \{0\}$. Therefore, it is not necessary to verify (as is done, for example, in the NTRU cryptosystem) whether an inverse exists for f in R_q.

According to the work [5], irreducibility of $P(x)$ in R_p is not required. In the case where $P(x)$ is not irreducible over R_p, it becomes necessary to verify the invertibility of the sampled polynomial f in R_p. However, as will be shown later, it is possible to select parameters such that the polynomial $P(x)$ is irreducible in both R_q and R_p. This enables uniform sampling of the polynomial f from the distribution $\mathcal{U}(R_p)$.

The NTWR Prime problem combines the NTRU Prime and LWR problems. Similar to the NTRU Prime problem, an instance of the NTWR Prime problem has the form $h = f^{-1} \star g$, where $g \leftarrow \psi_1$ and $f \leftarrow \psi_2$. The distributions ψ_1 and ψ_2 are certain probability distributions. Unlike the standard NTRU Prime instance, here we do not assume that $\psi_1 = \psi_2$. Instead, we allow the distribution ψ_1 to originate from the LWR problem.

With the above in mind, we can define the NTRU Prime problem using the following three definitions:

Definition 21 (NTWR Prime Distribution - $W(\bar{s}, f)$)**.** *Let q be a prime number, $h, k, n \in \mathbb{N}\backslash\{0\}$, and let $P(x)$ be an irreducible polynomial of degree n. Furthermore, let $\bar{s} \in R_q^k$, and let f be an invertible element in R_q. A sample from the NTWR Prime distribution $W(\bar{s}, f)$ is given by:*

$$(A, \bar{b}) = (A, f^{-1} \star \lfloor A \star \bar{s} \rceil_p) \in R_q^{h\times k} \times R_q^h, \tag{6}$$

where $A \leftarrow \mathcal{U}(R_q^{h\times k})$ and $\mathcal{U}(X)$ denotes the uniform distribution over the set X.

Let ψ and ψ^* be some probability distributions over R_q.

Definition 22 (Decision Problem Decision NTWR Prime - $DNTWRP(\psi, \psi^*)$)**.** *Let $k, h \in \mathbb{N}\backslash\{0\}$. An instance of the decision problem $DNTWRP(\psi, \psi^*)$ is given by an unknown distribution $\mathcal{D}$, which is either the uniform distribution or the distribution $W(\bar{s}, f)$ for some $\bar{s} \leftarrow \psi^k$ and $f \leftarrow \psi^*$. The $DNTWRP(\psi, \psi^*)$ problem is to determine which case holds.*

Definition 23 (Search Problem Search NTWR Prime - $SNTWRP(\psi, \psi^*)$)**.** *Let $h, k \in \mathbb{N}\backslash\{0\}$. An instance of the search problem $SNTWRP(\psi, \psi^*)$ consists of recovering the vector $\bar{s} \in R_q^k$ and the polynomial $f \in R_q$, given a sample from the distribution $W(\bar{s}, f)$.*

3 Key generation

In the NTWR Prime cryptosystem, the private and public key pair is defined as follows:

$$k_{\text{priv}} = (f, \bar{s}) \in R_q \times R_q^k, \tag{7}$$

$$k_{\text{pub}} = (A, \bar{b}) = \left(A, f_q^{-1} \star \underbrace{\lfloor A \star \bar{s} \rceil_p}_{=g} \right) \in R_q^{h\times k} \times R_q^h, \tag{8}$$

where $h \in \mathbb{N}$ denotes the number of LWE samples, and $k \in \mathbb{N}$ is the module dimension, i.e., the number of coefficients in the vector $\bar{a}_i \in R_q^k$ for $i = 1, \ldots, h$, which constitute the rows of matrix A. The operation $\star$ denotes polynomial multiplication in R_q. Furthermore, in equation (8), the rounding function is defined as:

$$\lfloor x \rceil_p := \left\lfloor (x \bmod q) \cdot \frac{p}{q} \right\rceil \in \mathbb{Z}_p, \tag{9}$$

applied individually to the coefficients of polynomials in R_q. Furthermore, f_q^{-1} denotes the inverse of f in R_q, i.e., $f_q^{-1} \star f \equiv 1 \mod q$. The existence of the polynomial f_q^{-1} for every non-zero f is guaranteed by the fact that R_q is choosen to be a Galois field.

The lengths of the private and public keys can be expressed as:

$$|k_{\text{priv}}| = (k+1) \cdot n \cdot \lceil \log_2 q \rceil, \tag{10}$$

$$|k_{\text{pub}}| = (h+1) \cdot k \cdot n \cdot \lceil \log_2 q \rceil. \tag{11}$$

Based on this, we calculate the maximum number of bytes (without encoding optimizations) required to store the matrix A and the vector $\bar{b}$.

At the current stage of development, the suggested distribution ψ^* is the uniform distribution $\mathcal{U}(R_p)$, while ψ is a bounded discrete Gaussian-like distribution, for instance a zero-centered binomial distribution $B_{R_{q,\beta}}$ with the number of trials equal β and the success probability for each trial $p = 1/2$ (in NTWR Prime, we set $\beta = 2$). So, $\Pr(-1) = \frac{1}{4} = \Pr(1)$ (where $-1 \equiv (q-1) \mod q$) and $\Pr(0) = \frac{1}{2}$ (the probability of remining integeres, in $\mathbb{Z}_q$, is zero). Thus, the coefficients of $\bar{s}$ are sampled from the $B^k_{R_{q,\beta}}$ distribution, meaning each coefficient of each polynomial from vector $\bar{s}$ is sampled from $B_{R_{q,\beta}}$. Furthermore, the coefficients of the matrix $A \in R_q^{h\times k}$ are sampled from the uniform distribution $\mathcal{U}(R_q^{h\times k})$.

The key generation process can be summarised as the Algorithm 1.

Algorithm 1 Key generation

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function KeyGen
  $\bar{s} \leftarrow B^k_{R_{q,\beta}}$
  $A \leftarrow \psi^{h\times k}_{R_q} = \mathcal{U}(R_q^{h\times k})$
  repeat
    $f \leftarrow \psi_{R_q}$
  until f_q^{-1} and f_p^{-1} exist
  $\bar{b} \leftarrow \lfloor A \star \bar{s} \rceil_p \cdot f_q^{-1} \mod q$
  return $k_{\text{pub}} = (A, \bar{b}), k_{\text{priv}} = (f, \bar{s})$
end function

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4 Encryption and decryption

To encrypt an N-bit message $m \in \{0,1\}^M$, we identify it with a polynomial in R_q whose coefficients correspond to the successive bits of m. Since the polynomials in R_q are of degree $n-1$ at most, where n is the degree of the polynomial $P(x)$, the condition $N \leq n$ holds. In what follows, we will consider a set of parameters with $n = 263$ and $n = 269$, which is sufficient to exchange (encapsulate) 256-bit keys as the secret messages being encrypted.

4.1 Encryption

The encryption process proceeds as follows:

– The $\bar{s'}$ from the distribution ψ^h_{enc}, where $\bar{s'} \in R^h_q$. The distrubution ψ^h_{enc} can be in practice choosen to be $B^h_{R_{q,\beta}}$ with, $\beta = 2$.
– Compute the first part of the ciphertext:

$$c_1 = (\langle \bar{s'}, \bar{b} \rangle + \left\lfloor \frac{q}{p} \right\rfloor m) \mod q \in R_q. \tag{12}$$

– Compute the second part of the ciphertext, treated as an element of R^k_q:

$$\bar{c_2} = \lfloor A^T \star \bar{s'} \rfloor_p \in R^k_q. \tag{13}$$

– The ciphertext is the pair $ct = (c_1, \bar{c_2}) \in R_q \times R^k_q$.

The encryption process can be summarised as the Algorithm 2.

Algorithm 2 Encryption

function ENCRYPT($message, k_{\text{pub}}$)
  $m \leftarrow toPoly(message)$
  $\bar{s}' \leftarrow B^h_{R_{q,\beta}}$
  $c_1 \leftarrow \bar{s}'^T \star \bar{b} + \left\lfloor \frac{q}{p} \right\rfloor m$
  $\bar{c}_2 \leftarrow \left\lfloor A^T \star \bar{s}' \right\rceil_p$
  return $(c_1, \bar{c}_2)$
end function

4.2 Decryption

For the decryption process, it is useful to introduce the function:

$$R_q \ni v := (c_1 \star f - \bar{c_2}^T \star \bar{s}) \mod q \tag{14}$$

Employing the expressions (12,13) and (8), we find that:

$$\begin{aligned} v &= \left[\left(\bar{s'}^T \bar{b} + \left\lfloor \frac{q}{p} \right\rfloor m \right) \mod q \cdot f - \left(\lfloor A^T \bar{s'} \rfloor_p \right)^T \bar{s} \right] \mod q \\ &= \left(\bar{s'}^T \lfloor A\bar{s} \rfloor_p \star f_q^{-1} \star f \right) \mod q - \left(\left\lfloor \bar{s'}^T A \right\rfloor_p \bar{s} \right) \mod q \\ &+ \left(\left\lfloor \frac{q}{p} \right\rfloor m \star f \right) \mod q \\ &= \left(\bar{s'}^T \lfloor A\bar{s} \rfloor_p - \left\lfloor \bar{s'}^T A \right\rfloor_p \bar{s} \right) \mod q + \left(\left\lfloor \frac{q}{p} \right\rfloor m \star f \right) \mod q, \end{aligned} \tag{15}$$

where the first bracket corresponds to the pure error contribution and the second bracket contains the message m to be extracted. For this purpose derive:

$$u = \left\lfloor v \frac{p}{q} \right\rceil \mod p \in R_p. \tag{16}$$

where $\lfloor x \rceil$ is the mathematically correct rounding:

$$\lfloor x \rceil = \begin{cases} \lfloor x \rfloor \text{ dla } x - \lfloor x \rfloor < 1/2 \\ \lceil x \rceil \text{ dla } x - \lfloor x \rfloor \geq 1/2 \end{cases}. \tag{17}$$

Then, we return the decrypted message as:

$$\tilde{m} = \left(u \star f_p^{-1}\right) \mod p, \tag{18}$$

where

$$f_p^{-1} \star f \equiv 1 \mod p,$$

which justifies the requirements of invertibility of f in R_p.

The decryption process can be summarised as the Algorithm 3.

Algorithm 3 Decryption

function DECRYPT($c_1, \bar{c}_2, k_{\text{priv}}, \bar{s}$)
  $v \leftarrow c_1 \star f - \bar{c}_2^T \star \bar{s}$
  $u \leftarrow \left\lfloor \frac{p}{q} v \right\rceil$
  $m \leftarrow u \star f_p^{-1}$
  return m
end function

4.3 Proof of decryption correctness

To prove the correctness of the decryption process, let us first note that $\left\lfloor v \cdot \frac{p}{q} \right\rceil \leq p$, where $v \in R_q$, always holds. The inequality is to be understood in terms of (all) coefficients of the polynomials.

Employing (15), the expression on (16), reads:

$$\begin{aligned} u &= \left\lfloor v \cdot \frac{p}{q} \right\rceil \\ &= \left\lfloor \frac{p}{q} \left(\bar{s'}^T \lfloor A\bar{s} \rceil_p - \left\lfloor \bar{s'}^T A \right\rceil_p \bar{s} \right) \mod q + \frac{p}{q} \left(\left\lfloor \frac{q}{p} \right\rceil m \star f \right) \mod q \right\rceil . \end{aligned} \tag{19}$$

The condition for u to let us, exactly, recover the message m is:

$$u = \left\lfloor v \cdot \frac{p}{q} \right\rceil = m \star f \mod p, \tag{20}$$

which can be written as:

$$v \cdot \frac{p}{q} = m \star f \mod p + r, \tag{21}$$

where $r \in (-1/2, 1/2)$. For $\left\lfloor v \cdot \frac{p}{q} \right\rceil < p$, we obtain the condition (for coefficients of the polynomials:

$$\left| v - \frac{q}{p} \left(m \star f \mod p\right) \right| < \frac{q}{2p}. \tag{22}$$

which guarantees correct decryption.

Using the expression for v, we get the condition:

$$\begin{aligned} &\left| \left(\bar{s'}^T \star \lfloor A \star \bar{s} \rceil_p - \left\lfloor \bar{s'}^T \star A \right\rceil_p \star \bar{s} \right) \mod q \right. \\ &\left. + \left(\left\lfloor \frac{q}{p} \right\rfloor m \star f \right) \mod q - \frac{q}{p} \left(m \star f \mod p\right) \right| < \frac{q}{2p}. \end{aligned} \tag{23}$$

Importantly, the difference of coefficients containing m can be upper-constrained, based on:

Lemma 1. *The following inequality holds:*

$$\frac{q}{p} \left(m \star f \mod p\right) - \left(\left\lfloor \frac{q}{p} \right\rfloor m \star f \right) \mod q < m \star f \mod q, \tag{24}$$

coefficient-wise.

Proof. Indeed, let us express each coefficient a of the polynomial $m \star f \in R_p$ as $a = kp + r_p$, where $k \in \mathbb{Z}$, $r_p < p$, and let $\frac{q}{p} = \left\lfloor \frac{q}{p} \right\rfloor + r$, where $r < 1$. Then in $\mathbb{Z}_q$:

$$\begin{aligned} &\frac{q}{p}(a \mod p) - \left(\left\lfloor \frac{q}{p} \right\rfloor a \right) \mod q = \\ &= \frac{q}{p} r_p - \left\lfloor \frac{q}{p} \right\rfloor kp \mod q - \left\lfloor \frac{q}{p} \right\rfloor r_p \mod q < \qquad (25) \\ &< (r_p + kp) \mod q = a \mod q \qquad (26) \end{aligned}$$

where inequality (25) follows from the estimate:

$$\frac{q}{p} r_p - \left\lfloor \frac{q}{p} \right\rfloor r_p \mod q = r_p \left(\frac{q}{p} - \left\lfloor \frac{q}{p} \right\rfloor \right) < r_p = r_p \mod q, \tag{27}$$

and equality (26) follows from:

$$\left\lfloor \frac{q}{p} \right\rfloor kp = \left(\left(\frac{q}{p} - r \right) kp \right) = \frac{q}{p} kp - rkp = kq - rkp. \tag{28}$$

Therefore,

$$\begin{aligned} &(r_p - \left\lfloor \frac{q}{p} \right\rfloor kp) \mod q = (r_p + kp) \mod q \\ &\qquad + (-kp - kq + rkp) \mod q < (r_p + kp) \mod q. \end{aligned} \tag{29}$$

QED

The stronger condition for the correction of decryption is therefore:

$$\left| \left(\bar{s'}^T \star \lfloor A \star \bar{s} \rfloor_p - \left\lfloor \bar{s'}^T \star A \right\rfloor_p \star \bar{s} \right) \mod q + m \star f \mod q \right| < \frac{q}{2p}. \tag{30}$$

Finally, using the triangle inequality, and taking the worst-case scenario with $m = 1$, we obtain the sufficient (strong) condition:

$$\left| \left(\bar{s'}^T \star \lfloor A \star \bar{s} \rfloor_p - \left\lfloor \bar{s'}^T \star A \right\rfloor_p \star \bar{s} \right) \mod q \right| + |1 \star f \mod q| < \frac{q}{2p}. \tag{31}$$

It is important to emphasize that inequality (31) is to be understood coordinate-wise over the polynomials. That is, it must hold for each coefficient separately. Consequently, as the degree n of the polynomials increases, the probability of correct decryption increases even if the sufficient condition is not strictly satisfied. This issue will be discussed in more detail in the following subsection.

4.4 Numerical study of the decryption correctness condition

To test the condition 31 let us consider the following cryptosystem parameters: $p = 2$, $q = 1907$, and uniform distributions over the interval $[0, 1]$, such that the coefficients of the polynomial f and the polynomial coefficients of the vectors $\bar{s}, \bar{s}'$ belong to $\mathbb{Z}_2$.

A total of 10,000 test trials were conducted for different (randomly generated) keys and messages. The analysis was performed for selected values of $n \in \{11, 53, 101, 257\} \subset \mathbb{P}$. The results of decryption correctness and the fulfillment of the sufficient condition (31) are presented in the Tab. 1.

n	Correct decryptions	Fulfillment of condition (31)	Avg. coeff. of non-fulfillment (31)
11	10000	1721	3 ± 2
53	10000	232	21 ± 8
101	10000	70	44 ± 13
257	10000	4	125 ± 16

Table 1: Numerical analysis of the condition (31).

As can be seen, in all cases, the message was correctly decrypted. This occurred despite the sufficient condition (31) not being satisfied. This condition is relatively strong, and as the presented example shows, its fulfillment is not necessary to achieve correct decryption.

It is worth noting that, as n increases, the number of cases in which condition (31) is satisfied decreases. Furthermore, on average, the failure to meet this condition results from discrepancies in fewer than half of the possible coefficients of the polynomial v.

To attempt to understand the above observations, let us recall the condition (31) for the case considered here, with $p = 2$:

$$\left| v - \frac{q}{2}\sigma \right| < \frac{q}{4}, \tag{32}$$

where $\sigma = (m \star f \bmod 2) \in \{0, 1\}$ for arbitrary m and f, understood in terms of the value of the polynomial coefficient. Furthermore, the possible values of the coefficients of the polynomial $v \in R_q$ belong to the set $\{0, 1, \ldots, q-1\}$. The total number of these coefficients is n.

As n increases, the number of partial conditions (for the polynomial coefficients) that must be simultaneously satisfied in order to meet condition (31) also increases.

Assuming that the satisfaction of each coefficient-level condition is an independent event, we should expect that the probability of satisfying the overall condition decreases as n grows. Specifically, let the probability of satisfying the condition for a single coefficient be $\eta < 1$. Then, the probability of satisfying the condition for all n coefficients is η^n, which is an exponentially decreasing function in n.

Let us now write inequality 32 for the case $\sigma = 0$:

$$|v| < \frac{q}{4}, \tag{33}$$

where, for the coefficients, $v \in \{0, 1, \ldots, q-1\}$. Therefore, the fraction of possible values of the coefficient v that satisfy the inequality is approximately equal to $1/4$.

Similarly, let us write inequality 32 for the case $\sigma = 1$:

$$\left| v - \frac{q}{2} \right| < \frac{q}{4}, \tag{34}$$

which can be rewritten as:

$$\frac{1}{4}q < v < \frac{3}{4}q. \tag{35}$$

Thus, the fraction of possible values of the coefficient v that satisfy the inequality is approximately equal to $1/2$.

Now consider the following scenario, which serves as a model for the case discussed here. Assume that the coefficient values v are uniformly distributed over the set $v \in \{0, 1, \ldots, q-1\}$. Also, assume that the value of σ takes values 0 and 1 with equal probability. Based on this, the probability of satisfying inequality 32 can be expressed as a combination of probabilities:

$$P = \frac{1}{2} \cdot \frac{1}{2} + \frac{1}{2} \cdot \frac{1}{4} = \frac{3}{8} \approx 0.38. \tag{36}$$

Therefore, we can expect that the average proportion of coefficients satisfying the inequalities is approximately 38%, which is in good agreement with the results presented in the table above regarding the average number of coefficients satisfying the sufficient condition.

5 Security analysis

The goal of this section is to demonstrate the minimal security level of the NTWR Prime cryptosystem. In this analysis, we refer to known results concerning the security of cryptosystems based on the LWR problem and the NTRU (Prime) cryptosystem. Next, we show that these problems can be attacked by having a solution to the NTWR Prime problem. We will carry out this analysis for both the decision and search problems.

5.1 Security of LWR

Despite its similarity to LWE, the security of the LWR problem is much less frequently analyzed in the literature. The main result comes from the original paper defining the LWR problem.

Theorem 1 ([12]). *For $q \leq pB2^{\lambda}$, the LWR problem is no easier than LWE with error size B (i.e., sampled from a B-bounded probability distribution over $\mathbb{Z}_q$) and security parameter λ, i.e., $q/p > 2^{\lambda}$ and $\alpha = E/q < 2^{-\lambda}$.*

The main limitation of the above theorem is the assumption of an exponential dependency of the ratio q/p on the security parameter λ, which is most often not feasible in practice.

With a limited number of samples that the adversary can obtain from the oracle, we have the following result from [13], which no longer requires such a wide range of parameters.

Theorem 2 ([13]). *Let k, l, m, p, γ be positive, λ be the security parameter, and let $p_{\max}$ denote the largest prime divisor of q. Suppose that $p_{\max} \geq 2B\gamma kmp$ and $k \geq (l + \lambda + 1)\log q/\log(2\gamma) + 2\lambda$. Then the LWR problem with a secret of size k, parameters p, q and a query limit m is as hard as LWE with a secret of size l, errors e sampled from a B-bounded probability distribution over $\mathbb{Z}_q$ and query limit m.*

In the case of R-LWE in the search version, we have the following reduction.

Theorem 3 ([14]). *Let $q > 2pB$. Then any algorithm that solves the Search R-LWE problem (i.e., recovers the secret s from k samples of the form $(x, f_s(x))$ obtained from the R-LWE process) with probability at least ε solves the Search R-LWR problem (i.e., recovers the secret s from k samples of the form $(x, \lfloor g_s(x) \rceil_p)$ obtained from the R-LWR process) with probability at least $\frac{\varepsilon^2}{(1+2Bp/q)^{nk}}$, where errors e are sampled from a B-bounded, balanced probability distribution over R_q.*

The decision version of the R-LWE problem does not have a full security proof. Some partial results (for the so-called *Computational Learning With Rounding over Rings*, R-CLWR) come from [15].

5.2 Security of NTRU (Prime)

The security of the NTRU and NTRU Prime cryptosystems relies on their relation to the shortest vector (SVP) and closest vector (CVP) problems on lattices. The SVP problem on a lattice $\mathcal{L}$ involves finding a vector $v \neq (0, .., 0)$ such that the norm $|v|$ attains a minimal value. The SVP problem is NP-hard [3]. The CVP problem on a lattice $\mathcal{L}$ with some $w \in \mathcal{L}$ involves finding a $v \in \mathcal{L}$ such that the norm $|v - w|$ attains a minimal value. The problem is considered harder than SVP. We can introduce a lattice in the form:

$$\mathcal{L}_h = (a, b) \in \mathbb{Z}^{2N} : a \star h = b \mod q, \tag{37}$$

where h is the public key. By the definition of h, it holds that $f \star h = g \mod q$. Thus, the private key (f, g) is an element (vector) in the lattice $\mathcal{L}_h$.

Since $f \star h - g = qt$, with some integer vector t, we can decompose the vector (f, g) in the following way: $(f, g) = (f, -t)M$, where the matrix M:

$$M = \begin{pmatrix} I \ H \ 0 \ qI \end{pmatrix}. \tag{38}$$

The submatrix H is formed from cyclic permutations of the vector h. The rows of matrix M constitute basis vectors in the lattice $\mathcal{L}_h$.

The fastest currently known method of solving the SVP problem on the NTRU public lattice is the hybrid attack described in [16].

Moreover, it is possible to carry out an attack that enables decryption of the message m. Carrying out this attack reduces to a problem approximating CVP; however, the exact complexity level of this problem is not known.

5.3 Security of NTWR Prime

Theorem 4. *Assume there exists an algorithm W capable of solving the (search/decision) NTWR Prime problem with advantage ε. Then a single use of algorithm W, with negligible additional computation, yields a solution to the corresponding (search/decision) M-LWR problem with advantage ε.*

Proof. Assume we can solve the NTWR Prime problem. We want to solve the Search M-LWE problem, i.e., given the pair $(A, \bar{b}) = (A, \lfloor A \star \bar{s} \rceil_p)$ we want to recover the secret $\bar{s}$. We sample $f \leftarrow \psi^*$ and construct an instance of the Search NTWR Prime problem in the form $(A, \bar{b}') = (A, f^{-1} \star \lfloor A \star \bar{s} \rceil_p)$, where $\bar{b}' = f^{-1} \star \bar{b}$. From the solution to the NTWR Prime problem, we obtain the secret $\bar{s}$.

Similarly, for the Decision M-LWR problem, from the pair $(A, \bar{b})$ we construct an instance of the Decision NTWR Prime problem $(A, \bar{b}')$, where $\bar{b}' = f^{-1} \star \bar{b}$ and using algorithm W with advantage ε, we distinguish whether the pair $(A, \bar{b}')$ comes from a uniform distribution or from the NTWR Prime distribution. If it comes from the latter, then the original pair comes from the M-LWR distribution.

QED

Theorem 5. *Assume there exists an algorithm W capable of solving the (search/decision) NTWR Prime problem with advantage ε. Then a single use of algorithm W, with negligible additional computation, yields a solution to the corresponding (search/decision) NTRU problem with advantage ε.*

Proof. Assume we can solve the NTWR Prime problem with $h = 1$. We want to solve the Search NTRU problem, i.e., given $b = f^{-1} \star g$ we want to recover the secret (f, g). We sample $f \leftarrow \psi^*$ and randomly choose a vector $\bar{a} \leftarrow \mathcal{U}(R_q^k)$ where $\mathcal{U}(X)$ denotes the uniform distribution over the set X. We then express

$$g = \lfloor \langle \bar{a}, \bar{s} \rangle \rfloor_p, \tag{39}$$

for some unknown vector $\bar{s}$. We thus obtain an instance of the Search NTWR Prime problem in the form $(\bar{a}, b) = (\bar{a}, f^{-1} \star \lfloor \langle \bar{a}, \bar{s} \rangle \rfloor_p)$. From solving the NTWR Prime problem, we get the secret pair $(f, \bar{s})$, which allows us to compute the NTRU secret key (f, g).

Similarly, for the Decision NTRU problem, we construct an instance of the Decision NTWR Prime problem $(\bar{a}, b)$ where $b = f^{-1} \star g$ and $g = \lfloor \langle \bar{a}, \bar{s} \rangle \rfloor_p$ and from algorithm W with advantage ε we distinguish whether the pair $(\bar{a}, b)$ comes from the uniform distribution or from the NTWR Prime distribution. If it comes from the latter, then the original pair comes from the NTRU distribution.

QED

From the above two theorems, it follows that the security of NTWR Prime is no less than the security of the weakest of the corresponding M-LWR and NTRU problems.

6 Parametres of the cryptosystem

The NTWR Prime cryptosystem is defined by the parameters listed in Table 3 and the distributions ψ and ψ^*, from which vectors $\bar{s} \in R_q^k$ and the polynomial $f \in \psi^*$ are sampled, respectively. At the current stage of work, the suggested distribution ψ^* is the uniform distribution $\mathcal{U}(R_p)$, whereas ψ is a β-bounded discrete Gaussian-like distribution (in NTWR Prime, we take $\beta = 2$). Additionally, the coefficients of matrix $A \in R_q^{h \times k}$ are sampled from the uniform distribution $\mathcal{U}(R_q^{h \times k})$.

n	Prime number defining the degree of the polynomial $P(x)$
p	Prime number defining $\mathbb{Z}_p$
q	Prime number defining $\mathbb{Z}_q$
h	Number of samples in the LWR process
k	Module dimension
β	Parameter of the β-bounded distribution

Table 2: Parameters of the NTWR Prime cryptosystem.

6.1 Requirements Regarding the NTRU Problem

According to the discussion presented in [17], the security criterion for the NTRU problem can be written as:

$$\|f\|, \|g\| \leq \sqrt{q}, \tag{40}$$

where $\|f\|$ denotes the Euclidean norm.

Since the polynomial f is sampled from the uniform distribution $\mathcal{U}(R_p)$, its coefficients are not greater than $p-1$. Therefore:

$$\max\|f\| = \sqrt{n}(p-1). \tag{41}$$

Hence, the condition $\|f\| \leq \sqrt{q}$ leads to:

$$\sqrt{n}(p-1) \leq \sqrt{q}. \tag{42}$$

In the NTWR Prime cryptosystem, we can write $g = \lfloor A \star \bar{s} \rceil_p \in R_q^h$. Therefore, for each $g_i \in R_q$, where $i = 1, ..., h$, we have:

$$\max\|g_i\| = \sqrt{n}(p-1). \tag{43}$$

We can thus state that the condition $\|g\| \leq \sqrt{q}$ also leads to expression (42).

The constraint on p, resulting from the NTRU security condition, can therefore be written as:

$$p \leq \sqrt{\frac{q}{n}} + 1. \tag{44}$$

6.2 Requirements Regarding the LWR Problem

Based on the results of the paper [18], the hardness criteria for the LWR problem can be written as:

$$p \geq \mathcal{O}(\sqrt{d \ln q}), \tag{45}$$

and

$$q \geq 2\beta dp. \tag{46}$$

In the case of the NTWR Prime problem, we assume $d = h \cdot k \cdot n$, which corresponds to the number of elements from $\mathbb{Z}_q$ in the public key.

6.3 Parameters for the NTWR Prime Problem

Below we assume $\beta = 2$, as is the case, for example, in the Kyber-768 and Kyber-1024 cryptosystems [19]. This choice stems from the fact that existing security proofs for the LWR problem rely on its reduction to the LWE problem. Therefore, we adopt the same β-bounded distribution parameter as in well-established LWE-based cryptosystems.

Additionally, we assume $h = 2 = k$, which corresponds to the LightSaber-KEM cryptosystem [20]. In general, we can take $h = k = 2, 3, 4, \ldots$, which requires further exploration.

Example parameter sets for the NTWR Prime cryptosystem are collected in Tab. 3. These parameters were derived according to the following methodology:

1. As the coefficient n, successive prime numbers greater than 256 were chosen.
2. For sufficiently large values of q, derived from estimates ensuring the security conditions for the NTRU and LWR problems, examples of prime numbers were found.
3. For selected prime numbers n and q, a lower bound for p (from the LWR problem) and an upper bound (from the NTRU problem) were determined.
4. For successive values of the parameters n and q, the irreducibility of the polynomial $P(x)$ over $\mathbb{Z}_q$ was analyzed.
5. In cases where the polynomial $P(x)$ was irreducible over $\mathbb{Z}_q$, the possible values of the parameter p were determined, consistent with the security conditions for the NTRU and LWR problems.
6. In cases where suitable values of the parameter p could be found, the irreducibility of the polynomial $P(x)$ over $\mathbb{Z}_p$ was analyzed.

The first two cases in Tab. 3 are included to illustrate situations where the security condition for one of the two problems (in this case, for LWR) is not satisfied, or the decryption error rate is high ($\sim 10^{-3}$). These two parameter sets are not recommended.

n	p	q	h	k	Length k_{pub} [B]	Error prob. of dec.	Sec. NTRU Prime	Sec. LWR	R_q-field	R_p-field
257	3	4540171	2	2	4369	$< 10^{-6}$	+	-	+	+
257	127	4540171	2	2	4369	$2 \cdot 10^{-3}$	+	+	+	-
263	157	$2 \cdot 10^9 + 33$	2	2	6312	$< 10^{-6}$	+	+	+	-
263	163	$2 \cdot 10^9 + 33$	2	2	6312	$< 10^{-6}$	+	+	+	-
269	163	$2 \cdot 10^9 + 11$	2	2	6456	$< 10^{-7}$	+	+	+	+
269	163	$2 \cdot 10^9 + 63$	2	2	6456	$< 10^{-7}$	+	+	+	+

Table 3: Example parameter sets for the NTWR Prime cryptosystem. The first two sets are not recommended.

For the NTWR Prime algorithm, it is recommended to use parameter sets for which both problems remain hard. Examples of such sets are provided in the remaining four parameter configurations. Moreover, in the last two cases, both R_q and R_p possess field structure.

7 Summary

In this article, we introduced a new post-quantum cryptographic asymmetric scheme that combines elements of NTRU Prime and the Learning With Rounding (LWR) problem. Building upon the NTWE cryptosystem by J. Gärtner [1], NTWR Prime replaces the Learning With Errors (LWE) mechanism with LWR to gain security and implementation benefits, such as determinism and reduced reliance on pseudorandomness. Additionally, the system is designed using a non-cyclotomic, irreducible polynomial with a symmetric Galois group, strengthening resistance against structural attacks.

We argued that NTWR Prime inherits the difficulty of both underlying hard problems–NTRU Prime and LWR–and thus offers a redundant layer of security suitable for high-assurance scenarios, albeit at the cost of efficiency and larger key sizes. In the article, we analysed the security assumptions, key generation, encryption/decryption procedures, and parameter choices. It concludes that NTWR Prime provides a robust and simplified alternative to existing lattice-based cryptosystems, with promising applications in contexts prioritizing maximal cryptographic resilience over performance.

At the current stage, we can formulate the following conclusions regarding the security of the proposed NTWR Prime cryptosystem:

- A minimal level of security has been proven for the NTWR Prime problem, on which the cryptosystem is based. This security is at least as strong as that of the NTRU Prime and LWR computational problems considered separately.
- Assuming that the proposed cryptosystem satisfies IND-CPA security (a formal proof remains to be completed), it will be possible to apply the Fujisaki–Okamoto transformation to construct an IND-CCA secure KEM based on the IND-CPA secure PKE scheme.
- The chosen parameters theoretically satisfy the proven security conditions for NTRU from the work [17].
- The chosen parameters also theoretically satisfy the proven security conditions for LWR as presented in [18]. These conditions were n ot met by the SABER cryptosystem.
- As shown, it is possible to select cryptosystem parameters in such a way that the security conditions for both NTRU and LWR problems are simultaneously satisfied.
- An irreducible polynomial $P(x)$ was chosen such that its Galois group has maximal order, in line with the security rationale for the NTRU Prime algorithm [5].
- As demonstrated, it is possible to choose parameters p, q, and n such that both R_q and R_p are finite fields. This simplifies the implementation of the cryptosystem by removing the need to verify the invertibility of the polynomial f in R_q and R_p. Therefore, a rejection sampling step is not required during private key generation, which provides additional resistance to side-channel attacks.
- The proposed cryptosystem offers higher "constructional security" compared to systems based on the LWE problem (e.g., Kyber or NTWE [1]), due to its deterministic algorithm. "Constructional security" refers to the simplicity of implementation and, in particular, the reduced likelihood of implementation-related errors. Except for the key generation stage, NTRU Prime does not require the use of randomness, whose generation is both time-consuming and a potential attack vector.

It is important to emphasize that further studies are needed to investigate relevant aspects of the cryptosystem that have not been discussed in this article. In particular, this concerns such aspects as:

- Analysis of decryption errors and execution times of the algorithm for low-level implementations of the cryptosystem. This may allow us to verify decryption correctness for error levels meeting the current postquantum standards.
- Analysis of the cases $h, k > 2$. In particular, selecting optimal values of h, k for security levels compliant with the NIST competition guidelines.
- Determination of security levels corresponding to the recommended parameter sets (in accordance with NIST competition guidelines).
- IND-CPA and IND-CCA1/2 security proofs.
- Optimization of cryptosystem parameters for performance while maintaining security.

We plan to discuss the above aspects in our forthcoming studies.

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Appendix: Toy example

To illustrate how the cryptosystem works, let us consider an example with $n = 2$, which leads to the polynomial:

$$P(x) = x^2 - x - 1. \tag{47}$$

Furthermore, we choose $p = 2$ and $q = 3$, so that the inequality $p < q$ is satisfied. Of course one has to keep in mind that neither the security conditions for the LWR problem nor for the NTRU Prime problem are satisfied for the values under consideration.

The prime numbers p and q allow us to define residue sets modulo $\mathbb{Z}_2$ and $\mathbb{Z}_3$. Consequently, the polynomial $P(x)$ can be expressed as:

$$P(x) = x^2 + x + 1 \text{ in } \mathbb{Z}_2, \tag{48}$$

$$P(x) = x^2 + 2x + 2 \text{ in } \mathbb{Z}_3. \tag{49}$$

One can now check that $P(0) = 1$ and $P(1) = 1$, which implies irreducibility of the polynomial in $\mathbb{Z}_2$, so R_2 is a finite field. Similarly, one can find that $P(0) = 2$, $P(1) = 2$ and $P(2) = 1$, which implies irreducibility of the polynomial in $\mathbb{Z}_3$, so R_3 is a finite field. Invertibility of all the polynomials in R_2 and R_3 is, therefore, guaranteed.

Furthermore, in accoradne with the discussion in Sec. 6, we choose $h = k = 2$.

7.1 Private Key

To generate the private key, we randomly choose polynomials $(f, \bar{s}) \in R_3 \times R_3^2$. Let us recall that the polynomial f is sampled from the uniform distribution $\mathcal{U}(R_p) = \mathcal{U}(R_2)$ and the vector $\bar{s}$ is sampled from the distribution $B^2_{R_3,2}$, so that $\Pr(0) = \frac{1}{2}$, $\Pr(1) = \frac{1}{4}$ and $\Pr(2) = \frac{1}{4}$. Here, the private key elements are choosen to be:

$$\begin{aligned} f &= 1 + x, \\ \bar{s} &= \begin{pmatrix} 1 + 2x \\ 2 + x \end{pmatrix}. \end{aligned} \tag{50}$$

7.2 Public Key

In the next step, we compute the key as:

$$(A, \bar{b}) = \left(A, \left(f^{-1} \star \lfloor A \star \bar{s} \rceil_2\right) \mod 3\right) \in R_3^{2\times 2} \times R_3^2, \tag{51}$$

where the matrix A is built from the transposed vectors $\bar{a}_i$ for $i = 1, 2$. The coefficients of the matrix A are sampled from the uniform distribution $\mathcal{U}(R_3^{2\times 2})$.

We randomly choose:

$$\bar{a}_1 = \begin{pmatrix} x \\ x + 1 \end{pmatrix}, \qquad \bar{a}_2 = \begin{pmatrix} 2 \\ 2x \end{pmatrix}, \tag{52}$$

which gives the first part of the public key:

$$A = \begin{pmatrix} (\ \bar{a}_1 \ ) \\ (\ \bar{a}_2 \ ) \end{pmatrix} = \begin{pmatrix} x & x + 1 \\ 2 & 2x \end{pmatrix}. \tag{53}$$

From this, we compute:

$$A \star \bar{s} = \begin{pmatrix} x & x + 1 \\ 2 & 2x \end{pmatrix} \star \begin{pmatrix} 1 + 2x \\ 2 + x \end{pmatrix} = \begin{pmatrix} 2 + x \\ 1 + x \end{pmatrix}. \tag{54}$$

Based on the definition of approximation, we compute:

$$\lfloor A \star \bar{s} \rceil_2 = \left\lfloor (A \star \bar{s} \mod 3) \cdot \frac{2}{3} \right\rceil = \left\lfloor \begin{pmatrix} 2 + x \\ 1 + x \end{pmatrix} \cdot \frac{2}{3} \right\rceil = \begin{pmatrix} 1 \\ 0 \end{pmatrix}. \tag{55}$$

Next, using the inverse $f^{-1} = 2x + 2$ in R_3, such that $f \star f^{-1} = 1 \mod 3$, we compute the second part of the public key:

$$\bar{b} = \left(f^{-1} \star \lfloor A \star \bar{s} \rceil_2\right) \mod 3 = \begin{pmatrix} 2 + 2x \\ 0 \end{pmatrix}. \tag{56}$$

7.3 Encryption

With the private and public keys, we can now proceed with the encryption process. In this case, four different messages can be encrypted: $m \in \{0,1\}^N = \{0, 1, x, 1+x\} \in R_2$.

To perform encryption, we randomly choose $\bar{s}' \in R_3^2$, employing the $B^2_{R_3,2}$ distribution, so that $\Pr(0) = \frac{1}{2}$, $\Pr(1) = \frac{1}{4}$ and $\Pr(2) = \frac{1}{4}$. distribution. Let:

$$\bar{s}' = \begin{pmatrix} x \\ 2+x \end{pmatrix}. \tag{57}$$

Then, the scalar part of the ciphertext is given by:

$$\begin{aligned} c_1 &= (\langle \bar{s}', \bar{b} \rangle + \left\lfloor \frac{3}{2} \right\rfloor m) \mod 3 \\ &= \begin{pmatrix} x & 2+x \end{pmatrix} \star \begin{pmatrix} 2+2x \\ 0 \end{pmatrix} \mod 3 + m \mod 3 \\ &= (2+x+m) \mod 3. \end{aligned}$$

The vector part of the ciphertext is:

$$\bar{c}_2 = \lfloor A^T \star \bar{s}' \rceil_2 = \left\lfloor \begin{pmatrix} x & 2 \\ 1+x & 2x \end{pmatrix} \star \begin{pmatrix} x \\ 2+x \end{pmatrix} \right\rceil_2 = \begin{pmatrix} 1 \\ x \end{pmatrix}.$$

The table below summarizes the possible ciphertext values $(c_1, \bar{c}_2)$:

m	c_1	$\bar{c}_2$
0	$2+x$	$\begin{pmatrix} 1 \\ x \end{pmatrix}$
1	x	$\begin{pmatrix} 1 \\ x \end{pmatrix}$
x	$2+2x$	$\begin{pmatrix} 1 \\ x \end{pmatrix}$
$1+x$	$2x$	$\begin{pmatrix} 1 \\ x \end{pmatrix}$

7.4 Decryption

The first step of the decryption process is computing:

$$v = (c_1 \star f - \langle \bar{c}_2, \bar{s} \rangle) \mod 3. \tag{58}$$

To compute v, we use:

$$\langle \bar{c}_2, \bar{s} \rangle = \begin{pmatrix} 1 & x \end{pmatrix} \star \begin{pmatrix} 1+2x \\ 2+x \end{pmatrix} = 2x + 2. \tag{59}$$

At this stage, we can verify the sufficient condition for correct decryption:

$$\left| v - \frac{3}{2} (m \star f \mod 2) \right| < \frac{3}{4}. \tag{60}$$

The results of computing v and verifying the sufficient condition are summarized in the table below:

m	c_1	$\bar{c}_2$	v	Condition		$\tilde{m}$
0	$2+x$	$\begin{pmatrix}1\\x\end{pmatrix}$	$1+2x$	$\lvert 2x+1 \rvert < \frac{3}{4}$	F	1
1	x	$\begin{pmatrix}1\\x\end{pmatrix}$	2	$\left\lvert \frac{1}{2} - \frac{3}{2}x \right\rvert < \frac{3}{4}$	F	x
x	$2+2x$	$\begin{pmatrix}1\\x\end{pmatrix}$	$2+x$	$\left\lvert \frac{1}{2} + x \right\rvert < \frac{3}{4}$	F	1
$1+x$	$2x$	$\begin{pmatrix}1\\x\end{pmatrix}$	$2x$	$\left\lvert \frac{1}{2}x \right\rvert < \frac{3}{4}$	T	$1+x$

As seen, in this example, only for the message $m = 1 + x$ the sufficient condition is met.

To complete the decryption, we compute $u = \left\lfloor v \cdot \frac{2}{3} \right\rceil \mod 2$, and then $\tilde{m} = \left(u \star f_2^{-1}\right) \mod 2$, where $f_2^{-1} = x$. As expected, in the case where the sufficient condition was met, the message is correctly decrypted. The low success rate of (25%) is due to the small value of the parameter q.

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